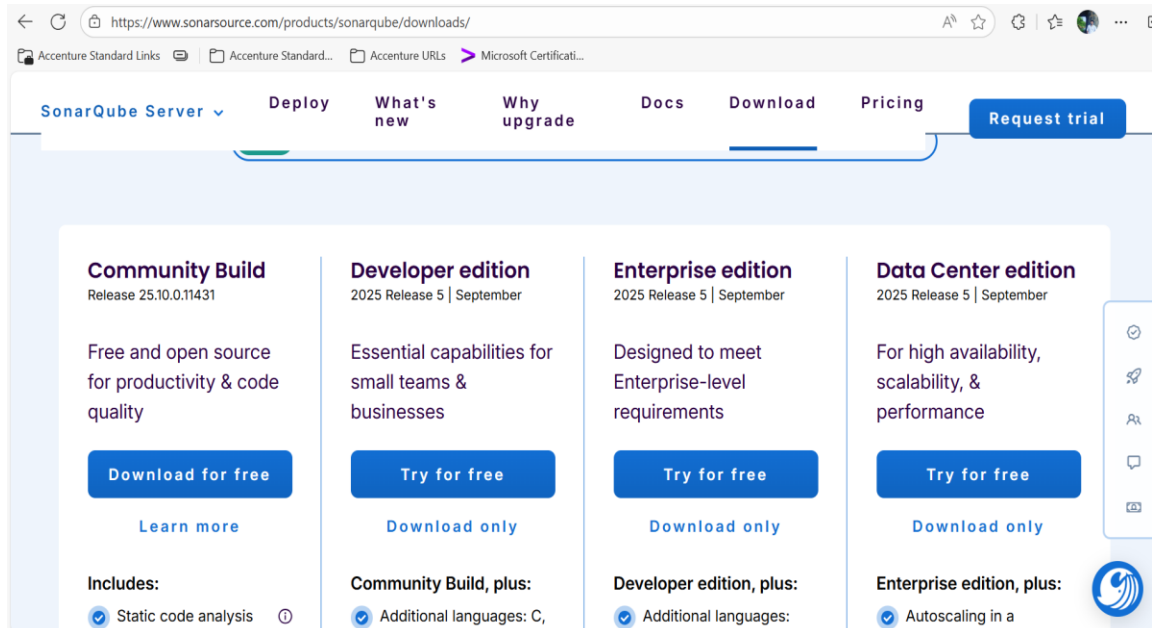


SonarQube - Activity

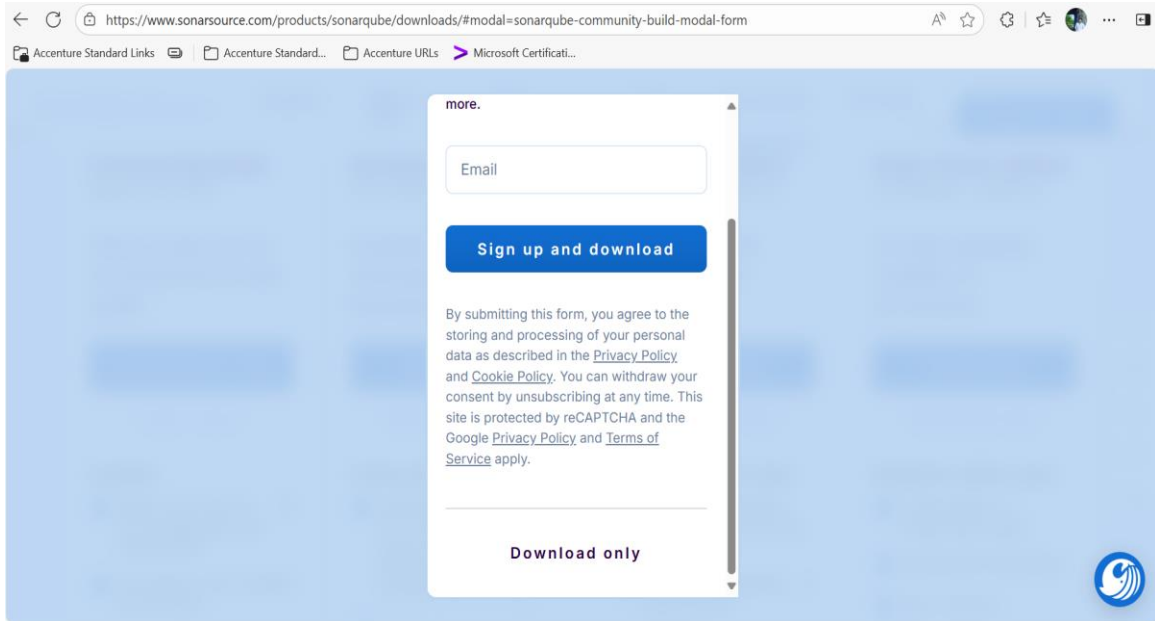
Activity - static code analysis with SonarQube and SonarScanner

1. Download SonarQube from the link below.

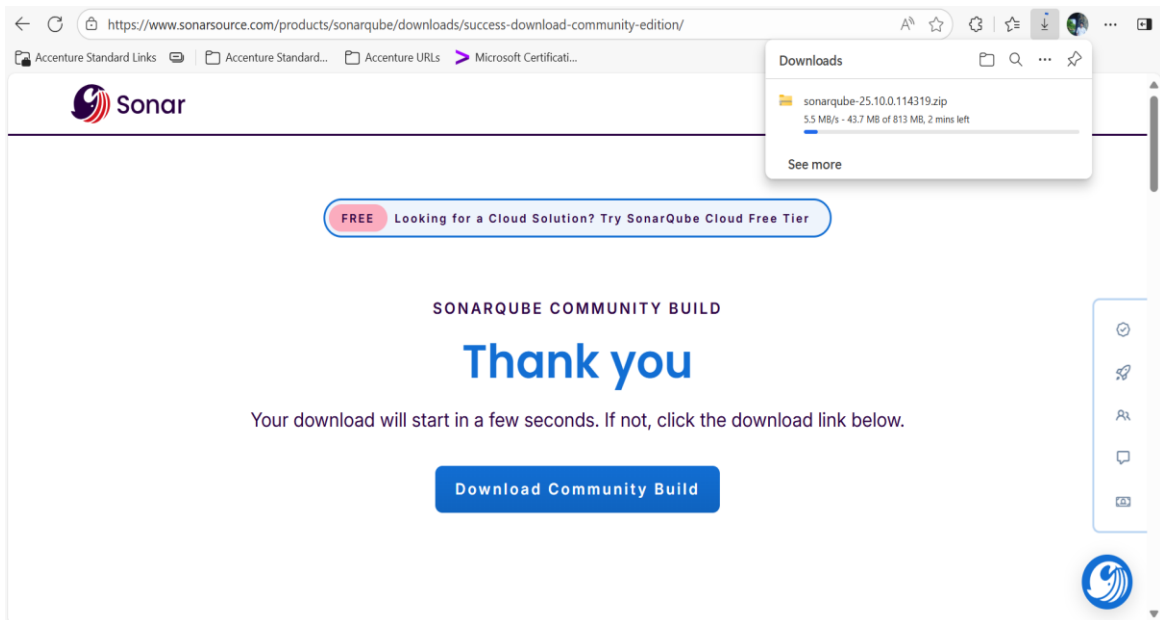
[Download SonarQube | Sonar](https://www.sonarsource.com/products/sonarqube/downloads/)



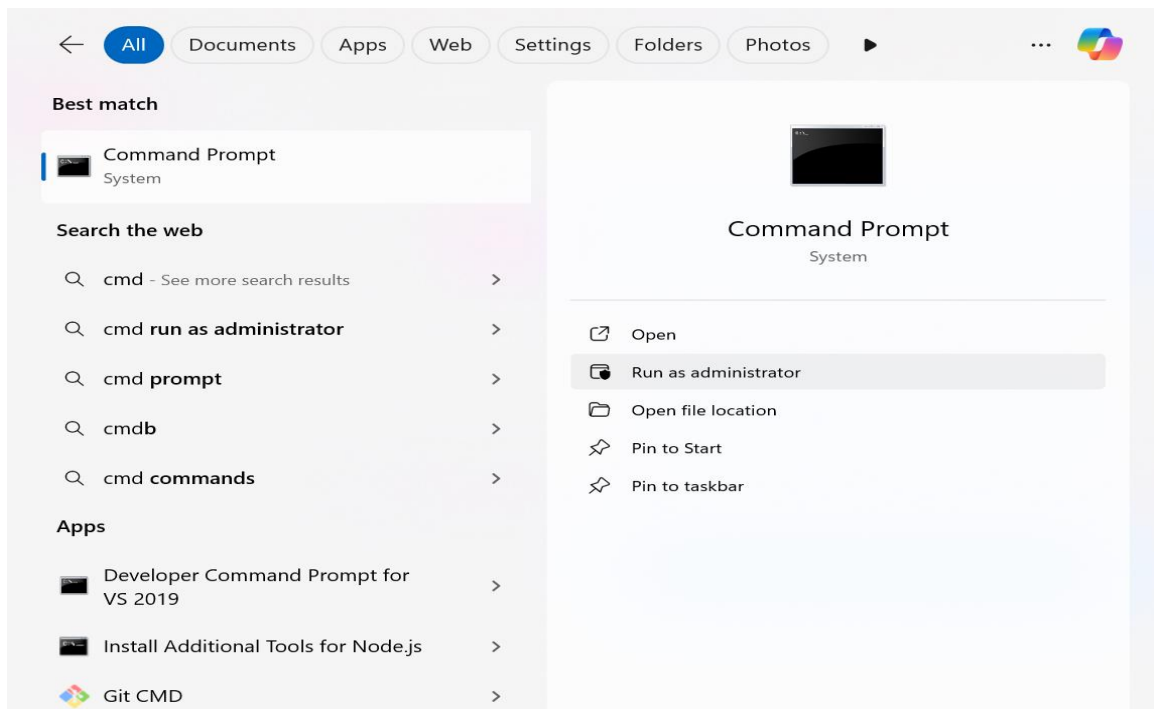
- **Download the Community Edition**, which is free, open-source, and designed to enhance productivity and code quality.
- **Subscribe to SonarQube updates** via email to stay informed about upcoming releases, product updates, and other news.
- **Subscription is optional** — you can still proceed with the download by scrolling down to the **Download Only** section.



2. Download will begin immediately, if not click on Download Community Build

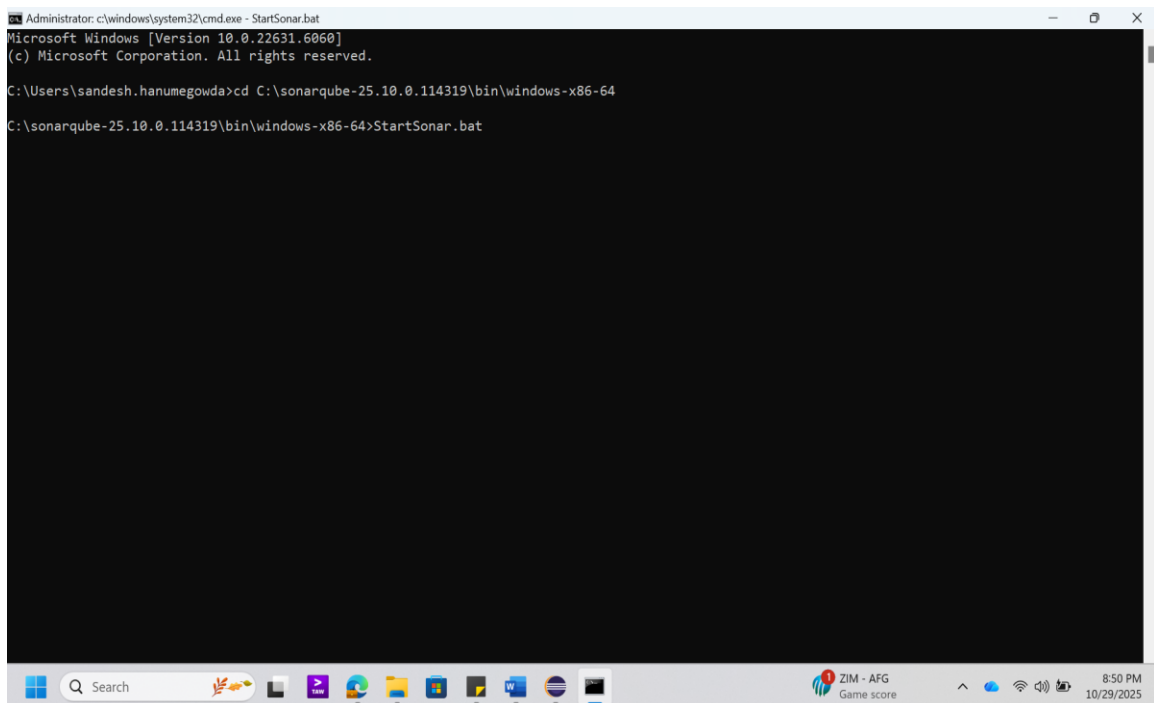


3. Once the Download is complete, extract the **sonarqube-25.10.0.114319.zip** file, then
4. Copy and paste the entire file **sonarqube-25.10.0.114319** into C:// drive
5. Open the command prompt with administrator access



6. Change the path to C:\sonarqube-25.10.0.114319\bin\windows-x86-64

7. And Start the SonarServer with StartSonar.bat command



Note: Wait for a while until the server is up and running.

```
Administrator: c:\windows\system32\cmd.exe - StartSonar.bat
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sandesh.hanumegowda>cd C:\sonarqube-25.10.0.114319\bin\windows-x86-64

C:\sonarqube-25.10.0.114319\bin\windows-x86-64>StartSonar.bat
Starting SonarQube...
2025.10.29 20:50:50 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory C:\sonarqube-25.10.0.114319\temp
2025.10.29 20:50:50 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:{}]
2025.10.29 20:50:50 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [C:\sonarqube-25.10.0.114319\elasticsearch]: C:\Program Files\Java\jdk-17\bin\java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=./bin/elasticsearch -Dcli.libs=lib/tools/server-cl
i -Des.path.home=C:\sonarqube-25.10.0.114319\elasticsearch -Des.path.conf=C:\sonarqube-25.10.0.114319\temp\conf\es -Des.distribution.type=tar -cp
C:\sonarqube-25.10.0.114319\elasticsearch\lib\*;C:\sonarqube-25.10.0.114319\elasticsearch\lib\cli-launcher\* org.elasticsearch.launcher.CliTool
launcher
2025.10.29 20:50:50 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
Standard Commons Logging discovery in action with spring-jcl: please remove commons-logging.jar from classpath in order to avoid potential confli
cts
2025.10.29 20:51:14 INFO app[[o.s.a.SchedulerImpl] Process[es] is up
2025.10.29 20:51:14 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[WEB_SERVER] from [C:\sonarqube-25.10.0.114319]: C:\Program Files\Java\j
dk-17\bin\java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=C:\sonarqube-25.10.0.114319\temp -XX:-OmitStackTraceInFastThrow --
add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java
.rmi/sun.rmi.transport=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=jav
a.base/sun.nio.ch=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UN
NAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost[127.*][:1] -cp ./lib/sonar-application-25.10.0.114319.jar
C:\sonarqube-25.10.0.114319\lib\jdbc\h2\h2-2.3.232.jar org.sonar.server.app.WebServer C:\sonarqube-25.10.0.114319\temp\sq-process50748455472870
42102properties
Standard Commons Logging discovery in action with spring-jcl: please remove commons-logging.jar from classpath in order to avoid potential confli
cts
```

Note: By default, the SonarQube server runs on **port 9000**.

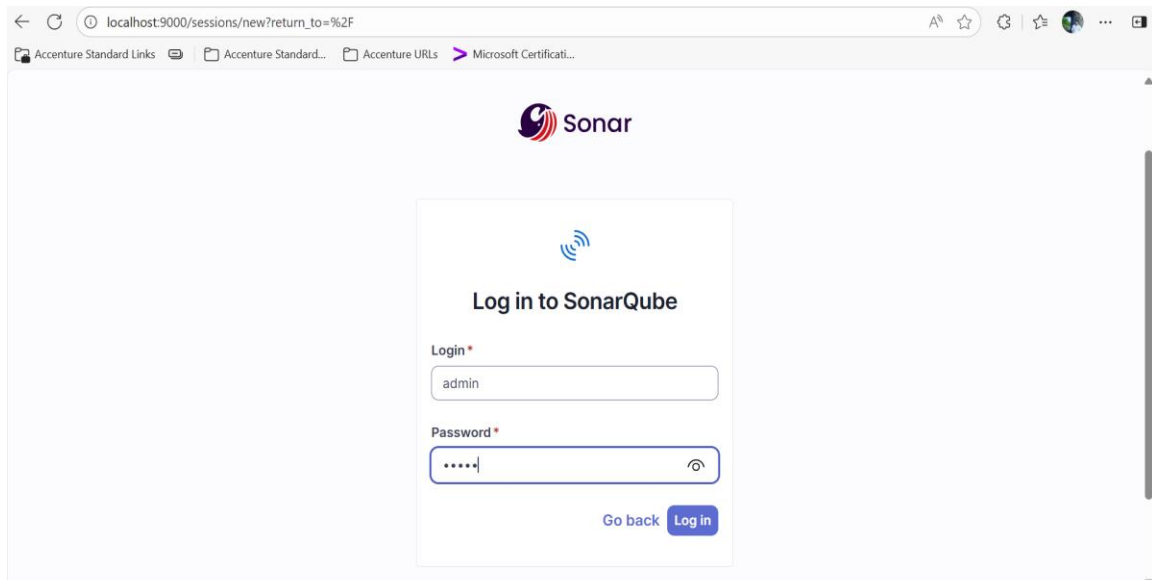
8. Use the following URL to access the SonarQube Dashboard:

<http://localhost:9000/>

Login credentials:

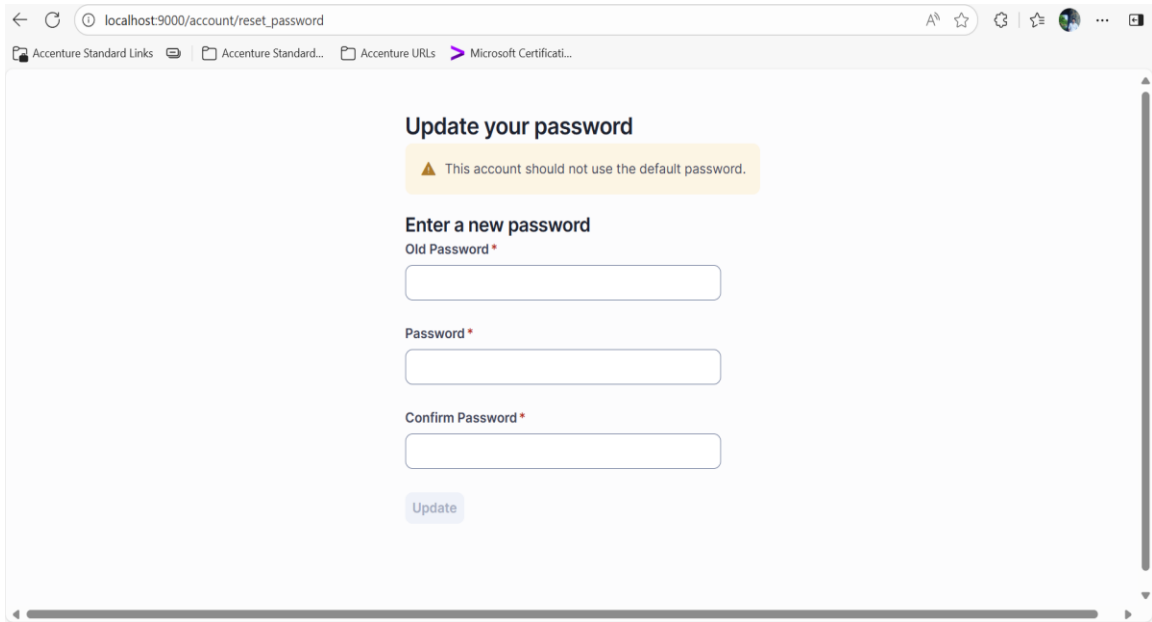
Username: admin

Password: admin



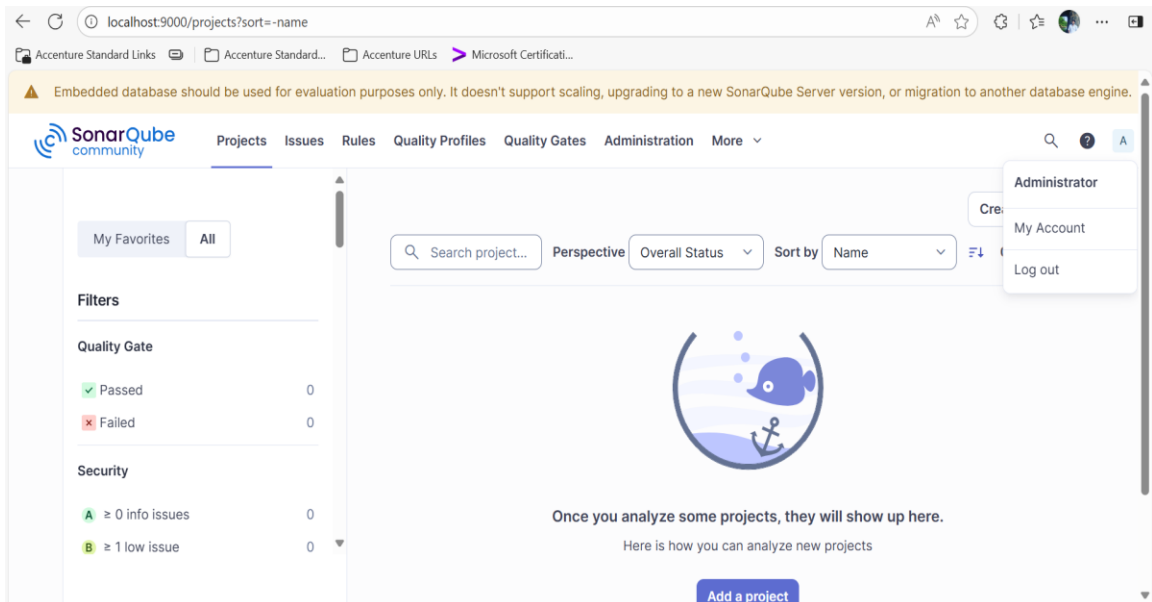
9. After logging onto, change the password:

Note: Please update the password and make a note of it for future reference.



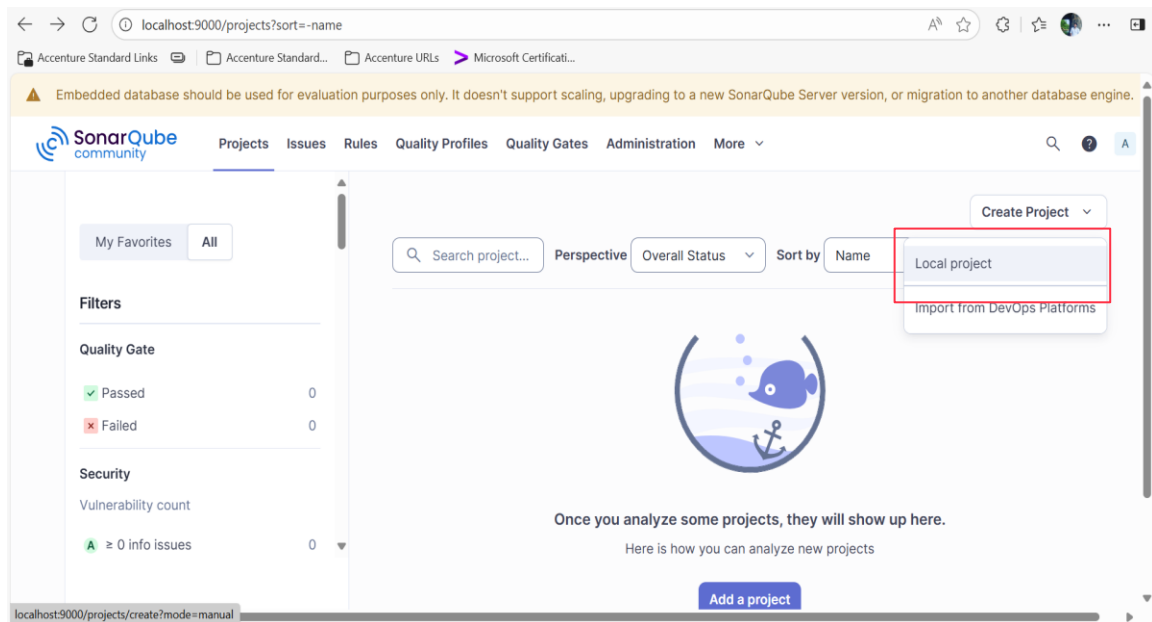
The screenshot shows a web browser window with the address bar displaying 'localhost:9000/account/reset_password'. The page title is 'Update your password'. A yellow warning box states: 'This account should not use the default password.' Below this, the section 'Enter a new password' contains three input fields: 'Old Password *', 'Password *', and 'Confirm Password *'. An 'Update' button is located at the bottom of the form.

10. After a successful login, you will be able to view the SonarQube Dashboard.



The screenshot shows the SonarQube Dashboard. The top navigation bar includes links for Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. A search bar and user profile dropdown (Administrator) are on the right. The left sidebar shows 'My Favorites' (All) and 'Filters' for Quality Gate (Passed: 0, Failed: 0) and Security (Info issues: 0, Low issue: 0). The main content area features a large blue fish icon and the text: 'Once you analyze some projects, they will show up here. Here is how you can analyze new projects'. An 'Add a project' button is at the bottom.

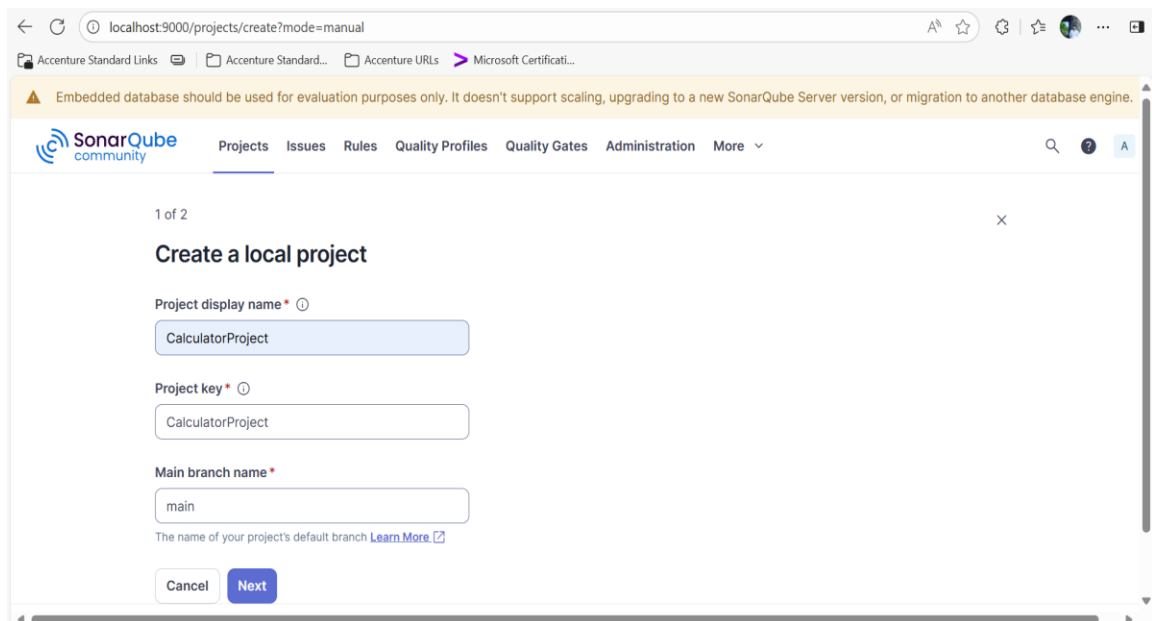
11. Click on **Create Project** → **Local Project**.



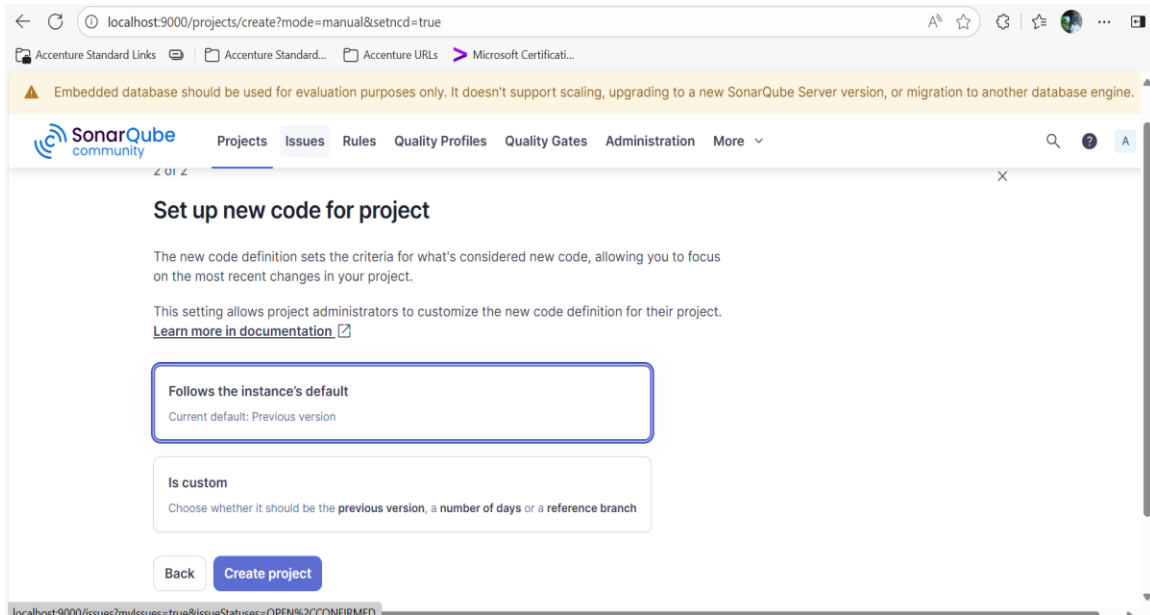
12. Fill the project details as per project requirement.

The **Project Display Name** will be automatically configured as the **Project Key** and the **Main Branch Name** should be set to **main**.

Once done, click **Next**.

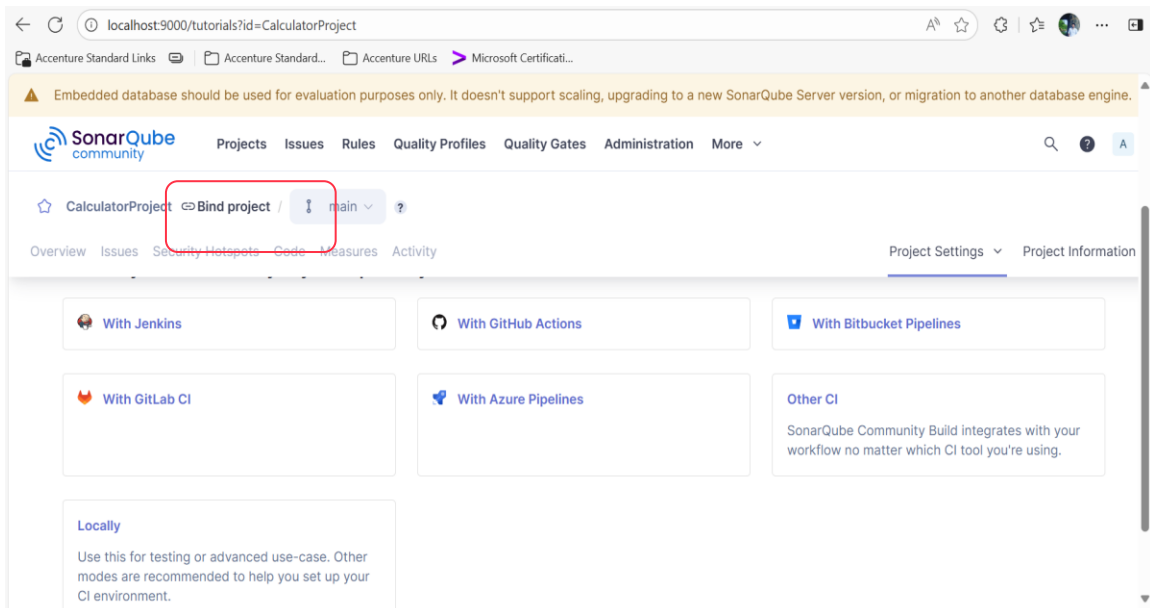


13. Use the **default settings** for new code in the project, then click on **Create Project**.

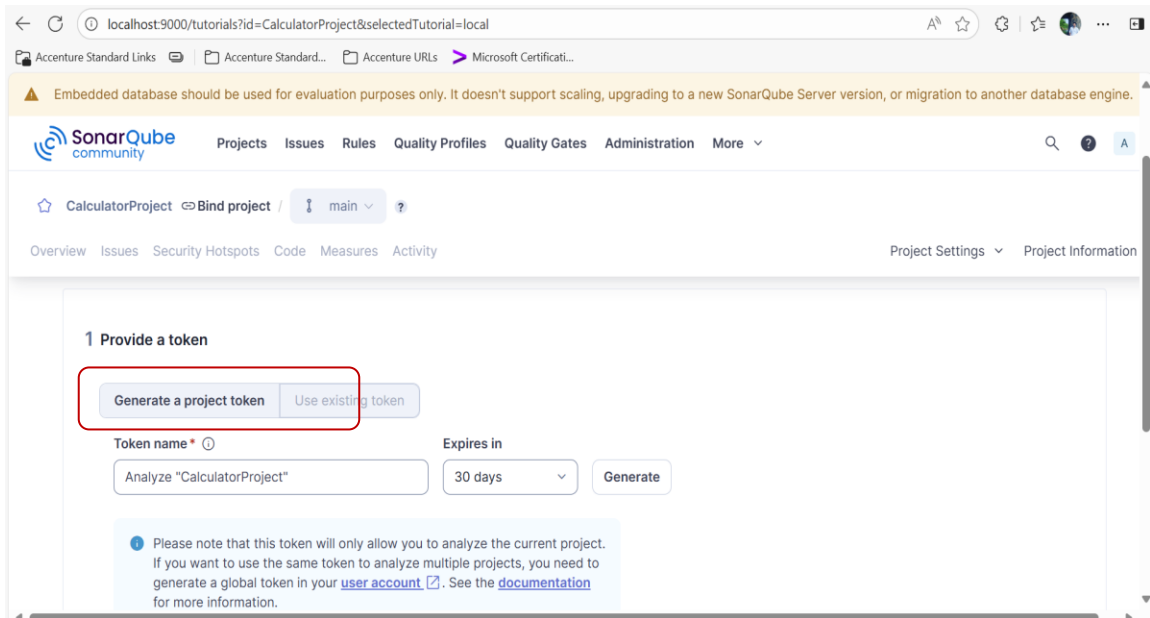


14. Once the project is created, you can proceed to **bind** the project.

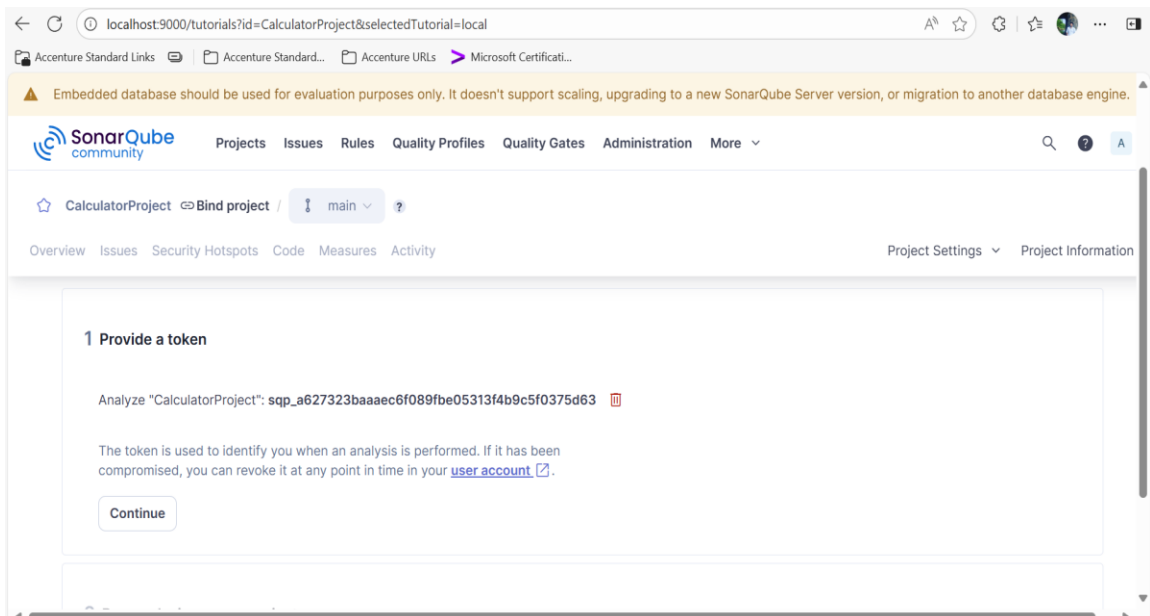
Note: Since we are running SonarQube locally, select the Locally option.



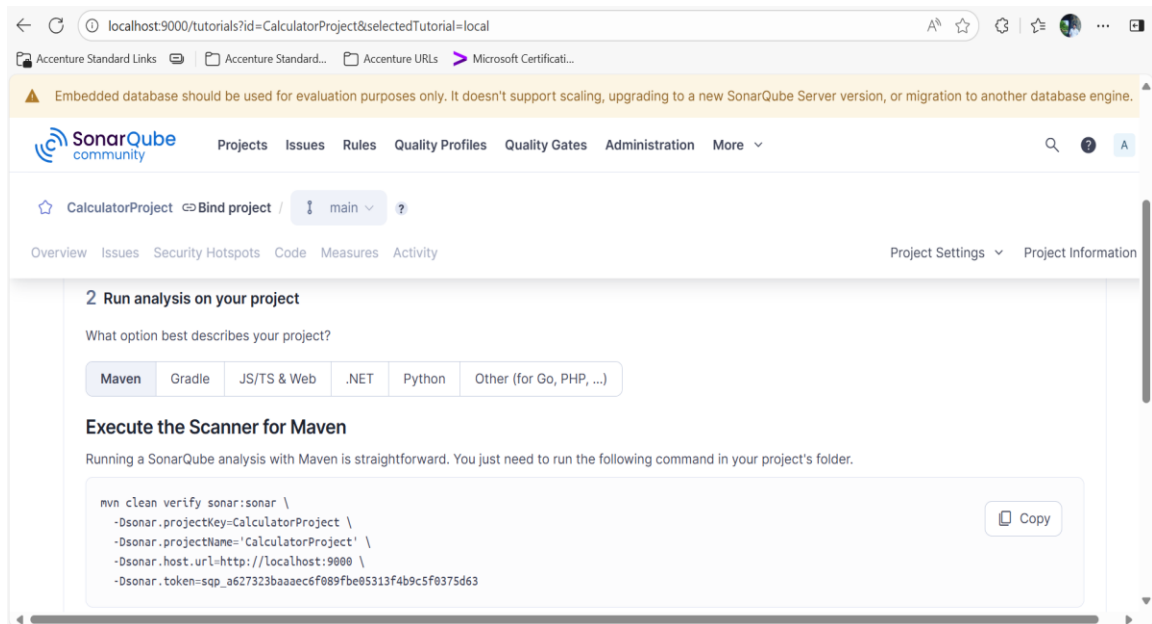
15. To bind the server project with eclipse demo for the analysis process, you need to **generate a token**. Click on **Generate Token** to proceed.



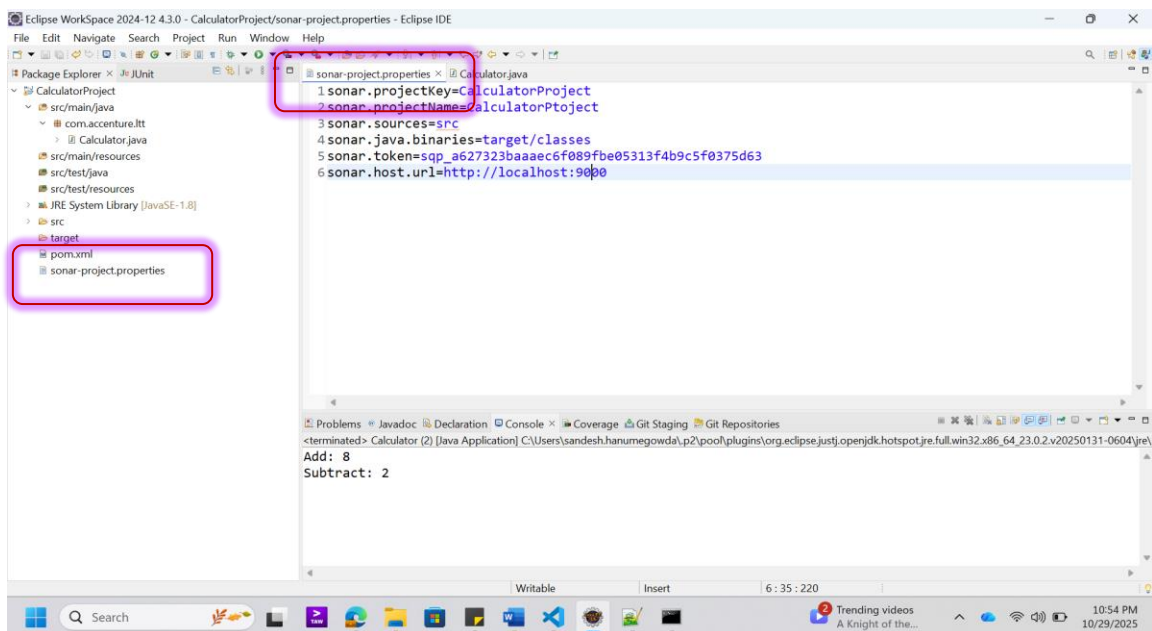
16. After the token is successfully generated, click on **Continue**.
Note: Make sure to save the token for future reference.



17. To run the analysis on your project, select **Maven** and copy the generated token as shown below.

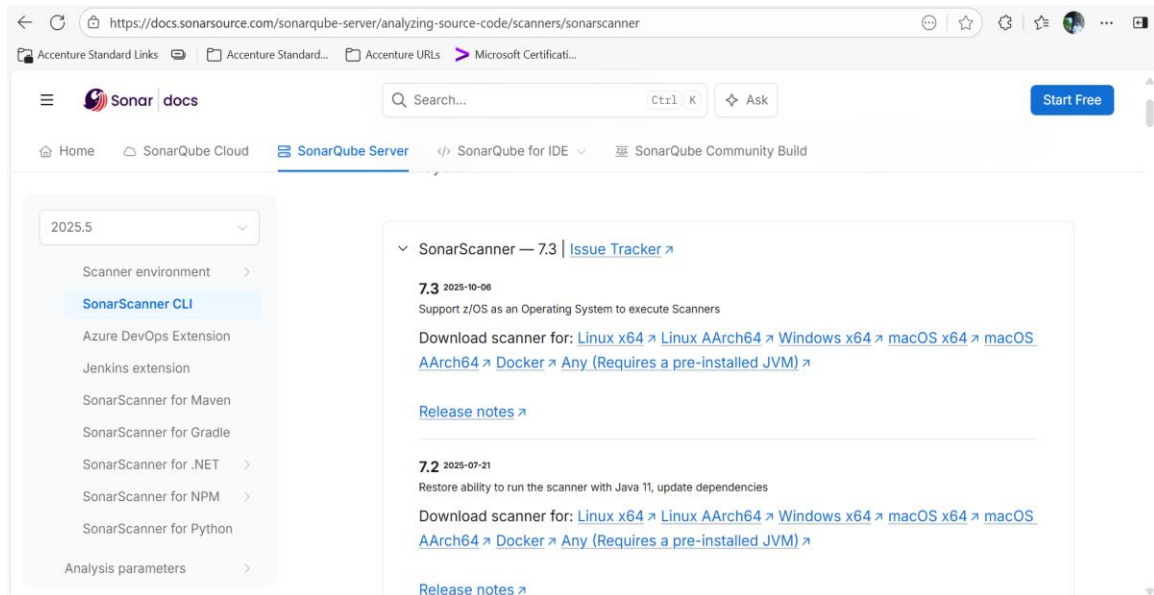


18. Open **Eclipse**, create or use existing maven project, and then add the copied properties by creating a file named **sonar-project.properties**.



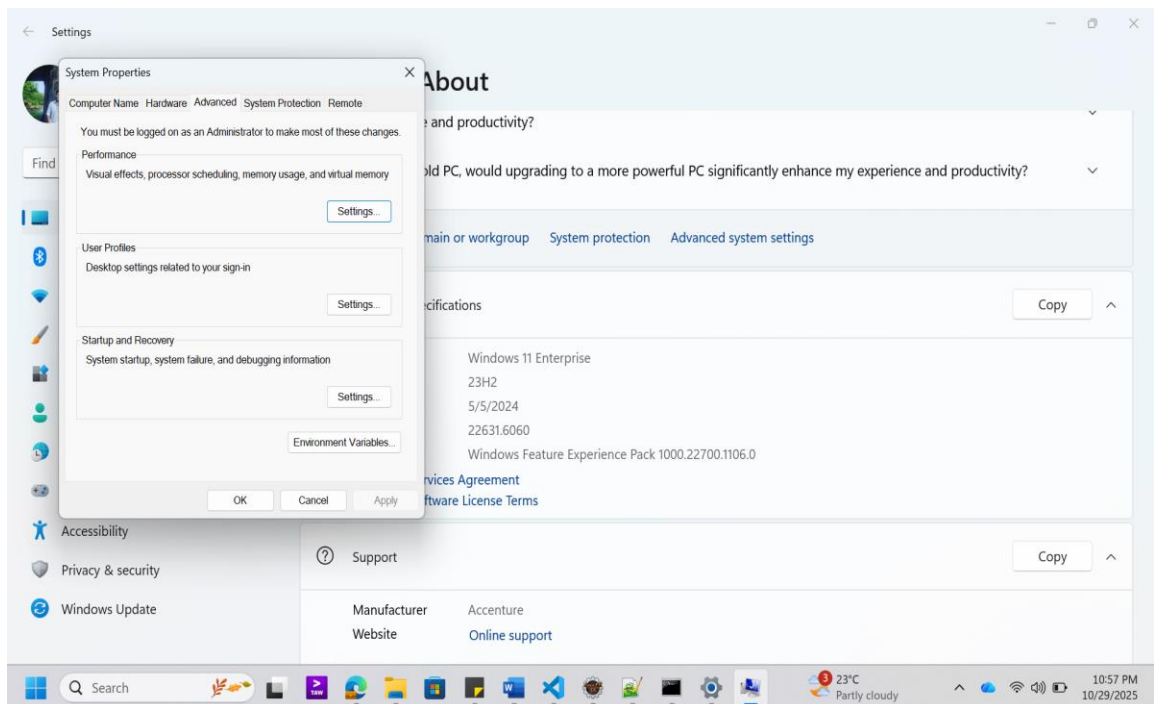
19. Download the **SonarScanner CLI** from the link provided below.

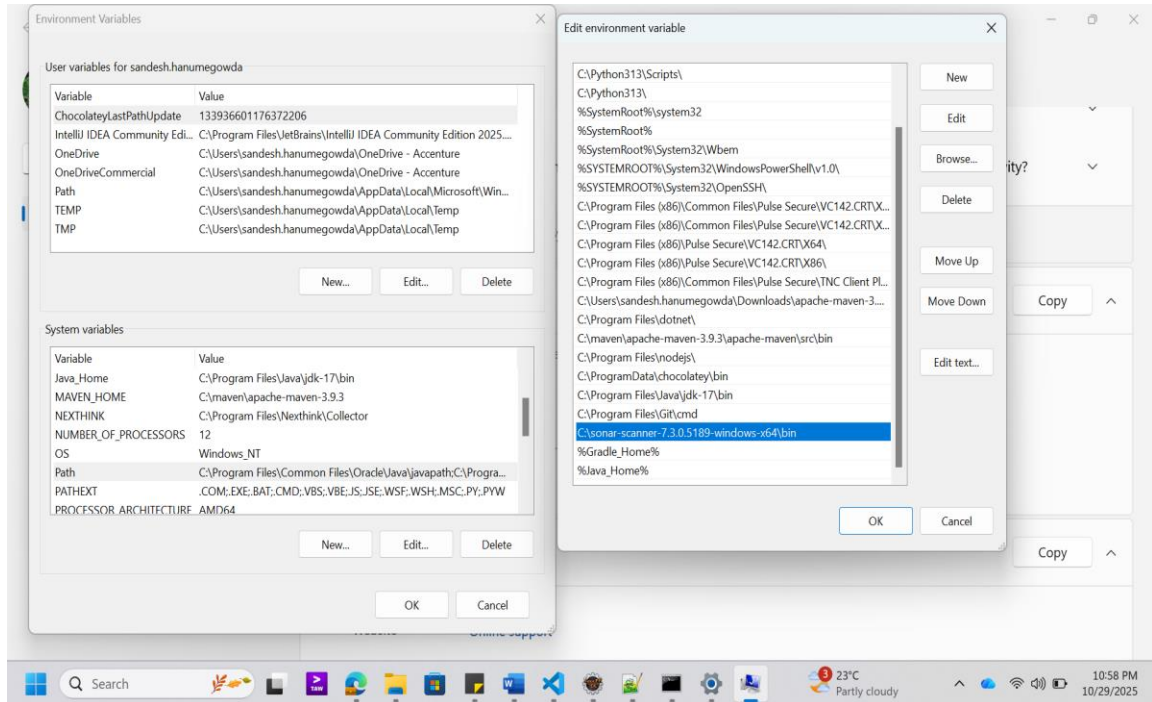
[SonarScanner CLI | Sonar Documentation](https://docs.sonarsource.com/sonarqube-server/analyzing-source-code/scanners/sonarscanner)



20. After downloading, **extract the file** and paste it into the **C:// drive**.

Ensure that **SonarScanner** is properly **configured in the System Variables** under **Environment Variables**.





21. Open the **Command Prompt (cmd)** and navigate to your **project directory**.

```

Command Prompt
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sandesh.hanumegowda>cd C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject
C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject>

```

22. Run the sonar-scanner

```
Command Prompt
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sandesh.hanumegowda>cd C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject

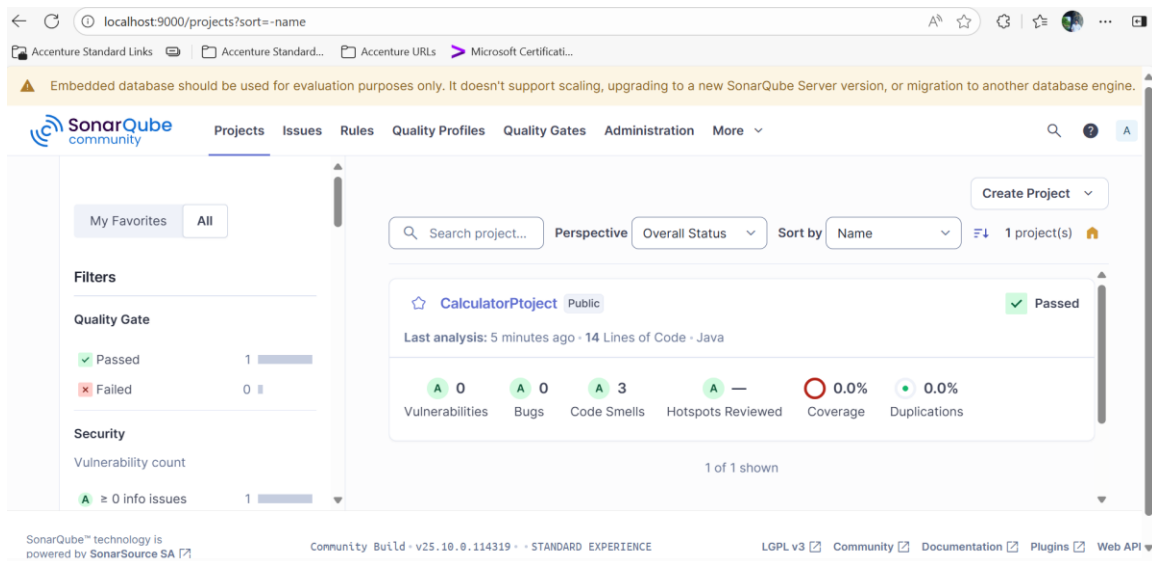
C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject>sonar-scanner
23:02:52.546 INFO Scanner configuration file: C:\sonar-scanner-7.3.0.5189-windows-x64\bin\..\conf\sonar-scanner.properties
23:02:52.576 INFO Project root configuration file: C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject\sonar-project.properties
23:02:52.614 INFO SonarScanner CLI 7.3.0.5189
23:02:52.617 INFO Windows 11 10.0 amd64
23:02:53.810 INFO Communicating with SonarQube Community Build 25.10.0.114319
23:02:53.811 INFO JRE provisioning: os[windows], arch[amd64]
23:02:55.251 INFO Starting SonarScanner Engine...
23:02:55.252 INFO Java 17.0.13 Eclipse Adoptium (64-bit)
23:02:56.934 INFO Load global settings
23:02:57.083 INFO Load global settings (done) | time=148ms
23:02:57.092 INFO Server id: 147B411E-AZowj0mUwhG3Lya7l3ZK
23:02:57.114 INFO Loading required plugins
23:02:57.115 INFO Load plugins index
23:02:57.136 INFO Load plugins index (done) | time=22ms
23:02:57.137 INFO Load/download plugins
23:02:57.711 INFO Load/download plugins (done) | time=574ms
23:02:58.662 INFO Process project properties
23:02:58.728 INFO Process project properties (done) | time=68ms
23:02:58.766 INFO Project key: CalculatorProject
23:02:58.768 INFO Base dir: C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject
23:02:58.772 INFO Working dir: C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject\.scannerwork
23:02:58.794 INFO Load project settings for component key: 'CalculatorProject'
23:02:58.835 INFO Load project settings for component key: 'CalculatorProject' (done) | time=40ms
23:02:58.912 INFO Load quality profiles
23:02:59.032 INFO Load quality profiles (done) | time=120ms
23:02:59.172 WARN SCM provider autodetection failed. Please use "sonar.scm.provider" to define SCM of your project, or disable the S
```

23. Observe the **execution completed successfully** message on the command prompt.

```
Command Prompt
23:04:08.068 INFO Using Git CLI to retrieve untracked files
23:04:08.464 WARN Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of inclusions specified via "sonar.text.inclusions"
23:04:08.465 INFO Starting the text and secrets analysis
23:04:08.468 INFO 1 source file to be analyzed for the text and secrets analysis
23:04:08.494 INFO 1/1 source file has been analyzed for the text and secrets analysis
23:04:08.497 INFO Sensor TextAndSecretsSensor [text] (done) | time=2631ms
23:04:08.506 INFO ----- Run sensors on project
23:04:08.944 INFO Sensor JavaProjectSensor [java]
23:04:08.953 INFO Sensor JavaProjectSensor [java] (done) | time=9ms
23:04:08.954 INFO Sensor Zero Coverage Sensor
23:04:08.968 INFO Sensor Zero Coverage Sensor (done) | time=14ms
23:04:08.969 INFO Sensor Java CPD Block Indexer
23:04:09.013 INFO Sensor Java CPD Block Indexer (done) | time=45ms
23:04:09.014 INFO ----- Gather SCA dependencies on project
23:04:09.020 INFO Dependency analysis skipped
23:04:09.024 INFO SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
23:04:09.027 INFO CPD Executor 1 file had no CPD blocks
23:04:09.028 INFO CPD Executor Calculating CPD for 0 files
23:04:09.029 INFO CPD Executor CPD calculation finished (done) | time=0ms
23:04:09.529 INFO Analysis report generated in 257ms, dir size=260.5 kB
23:04:09.688 INFO Analysis report compressed in 117ms, zip size=28.7 kB
23:04:09.792 INFO Analysis report uploaded in 103ms
23:04:09.799 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=CalculatorProject
23:04:09.800 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
23:04:09.802 INFO More about the report processing at http://localhost:9000/api/ce/task?id=13082dae-c50d-45de-9254-539b06a39677
23:04:09.842 INFO Analysis total time: 12.143 s
23:04:09.846 INFO SonarScanner Engine completed successfully
23:04:09.974 INFO EXECUTION SUCCESS
23:04:09.976 INFO Total time: 16.495s

C:\Users\sandesh.hanumegowda\Downloads\Eclipse Workspace 2024-12 4.3.0\CalculatorProject>|
```

24. Log in to <http://localhost:9000/> (**SonarQube Server**) to view the analysis report.



25. Configured and executed SonarQube locally for project code analysis using SonarScanner integrated with Eclipse and Maven.

26. The analysis results were verified via the SonarQube Dashboard at <http://localhost:9000/>